AUTO TICKET CLASSIFICATION – OVERVIEW AND PROBLEM STATEMENTS

1. Project Overview

Auto Ticket Classification is a machine learning and Natural Language Processing (NLP) project that automatically classifies support tickets into predefined categories based on their content.

In many organizations, customers and employees raise support tickets for issues related to technical problems, billing, account access, network failures, software errors, and other services. Traditionally, support teams manually read each ticket and assign it to the appropriate department.

This manual process takes time, increases workload, and may result in incorrect categorization or delays in resolving issues.

The Auto Ticket Classification system solves this problem by analyzing the text of a ticket and automatically assigning it to the appropriate category or support team. It can also be extended to predict ticket priority, identify urgency, and recommend the appropriate department.

Example:

Support Ticket	Predicted Category
I cannot log in to my account.	Account Access
My internet connection is not working.	Network Issue
I was charged twice for my subscription.	Billing Issue
The application crashes when I open it.	Software Issue
I need to reset my password.	Password Reset

2. Problem Statement

Organizations receive a large volume of support tickets every day from customers and employees. Manually reviewing, categorizing, and assigning these tickets is time-consuming and inefficient. Incorrect classification can lead to tickets being routed to the wrong department, increasing response times and reducing customer satisfaction.

Therefore, there is a need to develop an automated ticket classification system using Machine Learning and Natural Language Processing (NLP) that analyzes ticket descriptions and automatically assigns them to predefined categories.

The proposed system aims to reduce manual effort, improve classification accuracy, speed up ticket routing, and help support teams manage tickets more efficiently.

3. Specific Problem Statements

The project can address the following problems:

Manual Ticket Classification: Support teams spend considerable time reading and categorizing tickets manually.

Incorrect Categorization: Tickets may be assigned to the wrong category or department, resulting in unnecessary delays.

High Volume of Tickets: Organizations may receive thousands of tickets, making manual classification difficult to scale.

Delayed Ticket Resolution: Incorrect routing and manual processing increase the time required to resolve customer issues.

Inconsistent Classification: Different support agents may assign different categories to similar tickets.

Lack of Automatic Prioritization: Urgent issues may not be identified quickly enough, affecting service quality.

4. Objectives of the Project

The main objectives of the Auto Ticket Classification project are:

To automate the classification of support tickets.

To apply NLP techniques to understand ticket descriptions.

To train a machine learning model using historical ticket data.

To classify tickets into predefined categories, such as technical issues, billing, network problems, and account access.

To reduce manual effort and minimize classification errors.

To improve ticket routing and overall support team efficiency.

To evaluate the model using metrics such as accuracy, precision, recall, and F1-score.

5. Proposed System

The proposed system automatically processes incoming support tickets and predicts their appropriate categories.

System workflow:

Ticket Input: The user submits a support ticket containing a subject and description.

Text Preprocessing: The system cleans the text by removing unnecessary characters and handling text normalization.

Feature Extraction: NLP techniques convert the text into numerical features using methods such as TF-IDF.

Model Prediction: A trained machine learning algorithm analyzes the features and predicts the ticket category.

Ticket Classification: The system assigns the predicted category to the ticket.

Ticket Routing: Optionally, the ticket is forwarded to the appropriate support team.

Performance Evaluation: The model's predictions are evaluated against known ticket categories.

Technologies that can be used:

Component	Suggested Technologies
Programming Language	Python
Data Processing	Pandas, NumPy
Natural Language Processing	NLTK, spaCy
Feature Extraction	TF-IDF
Machine Learning	Naive Bayes, Logistic Regression, Support Vector Machine
Deep Learning (Optional)	LSTM, BERT
Model Evaluation	Scikit-learn
User Interface (Optional)	Flask, Streamlit
Database (Optional)	MySQL, SQLite

6. Expected Outcomes

The expected outcomes of this project include:

Automatic categorization of incoming support tickets.

Faster ticket assignment to the appropriate departments.

Reduced manual workload for support agents.

More consistent ticket classification.

Improved support team productivity.

A scalable approach to handling large volumes of tickets.

The actual classification performance will depend on the quality, quantity, and balance of the training data.

7. Short Problem Statement for Project Submission

If you need a concise version for a college project report or presentation, you can use the following:

Problem Statement:

The increasing volume of customer support tickets makes manual classification time-consuming, inefficient, and prone to errors. Incorrect categorization can delay ticket assignment and issue resolution. This project aims to develop an automated ticket classification system using Natural Language Processing and Machine Learning techniques to analyze ticket descriptions and assign them to appropriate categories. The system is intended to reduce manual effort, improve classification accuracy, and enhance the efficiency of support ticket management.

8. Conclusion

Auto Ticket Classification is an effective application of machine learning and NLP that automates the categorization of support tickets. By predicting ticket categories from textual descriptions, the system can reduce manual work, improve ticket routing, and help organizations manage customer support requests more efficiently.

Recommended starting point: For a student-level project, begin with Python, TF-IDF feature extraction, and a Logistic Regression or Naive Bayes classifier. Once the basic model works, you can extend it with automatic ticket priority prediction and a web-based interface.